

THE HODGE CONJECTURE: AN ATTEMPT IN FULL, AND THE STATEMENTS AT WHICH IT STOPS

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ABSTRACT. We follow every route to the Hodge conjecture that the present literature makes available, prove each reduction that can be proved, and stop at the first statement on each route that cannot. The Hodge conjecture is not proved here. Every Hodge class of degree two, of codegree two, or on a variety of dimension at most three is algebraic, and the conjecture descends along surjective morphisms. By the author's earlier work it is equivalent to two statements: the conjecture for abelian varieties, and the conjecture modulo abelian varieties. For abelian varieties of Weil type the problem comes down to one class on a general member of a family, and we show that the cycles Schoen constructed on Prym varieties of triple covers give, on each such Prym variety, an explicit subvariety of half dimension whose class is a positive multiple of the polarization power plus a nonzero Weil class. If that subvariety were semiregular, the Weil classes would be algebraic on every member of the split Weil families of $\mathbb{Q}(\sqrt{-3})$ in every dimension. We cannot decide whether it is. We also show that every abelian variety of Weil type for $\mathbb{Q}(\sqrt{-3})$ is, up to isogeny, a factor of such a Prym variety, so that its Weil classes are algebraic exactly when those of the complementary factor are; a count of dimensions indicates that this cannot reach the general member in dimension eight or more. For the remaining classes on abelian varieties, and for varieties that are not of abelian type, we record what is proved and locate the first open case of each: the exceptional classes on the square of a Mumford fourfold, and the square of a K3 surface with real multiplication.

1. INTRODUCTION

Let X be a smooth complex projective variety of dimension n . A class in $H^{2p}(X, \mathbb{Q})$ is a *Hodge class* if its image in $H^{2p}(X, \mathbb{C})$ lies in $H^{p,p}(X)$, and it is *algebraic* if it is a rational combination of classes of subvarieties of codimension p . Algebraic classes are Hodge classes. The Hodge conjecture says that the converse holds.

Conjecture 1.1 (Hodge). *Every Hodge class on a smooth complex projective variety is algebraic.*

This paper is an attempt at the conjecture in full. By that we mean something specific. We take every route that the literature makes available, push it as far as it can be pushed with complete proofs, and name the first statement on it that we cannot prove. We do not prove the conjecture, and nothing here should be read as a claim that it is close to proved. What the attempt produces is a map: a short list of statements, each open, each precise, from which the conjecture would follow, together with an account of why the obvious ways of proving each of them fail.

Section 2 proves the reductions that hold for every variety: degree two, codegree two, dimension at most three, descent along surjective morphisms, and the reduction of every degree to the middle one under the Lefschetz standard conjecture. Section 3 recalls from [BhH8] that, beyond these, the conjecture is equivalent to two statements, the conjecture for abelian varieties and the conjecture modulo abelian varieties, and lists the other minimal sets of hypotheses from which it follows. Sections 4 and 5 attack abelian varieties. Section 5 contains the main new theorem of the paper, Theorem 5.2, which turns Schoen's cycles into a single explicit subvariety

of the right shape for a deformation argument, and isolates the question on which the argument turns. Section 6 treats the classes on abelian varieties that divisors and Weil classes do not generate, and Section 7 the varieties that are not of abelian type. Section 8 tries to spread Schoen's construction beyond the Prym varieties. It proves that every abelian variety of Weil type for $\mathbb{Q}(\sqrt{-3})$ is, up to isogeny, a factor of the Prym variety of an étale triple cover, so that its Weil classes are algebraic exactly when those of the complementary factor are (Theorem 8.2); it shows by a count why this cannot by itself reach the general member once the dimension is at least eight; and it explains why the construction, carried to other varieties, leads back to (F2) and (F3'). Section 9 collects what is proved and what is not.

Table 1 summarises where each route stops.

Route	What is proved	Where it stops
Divisors and hard Lefschetz	degrees 2 and $2n - 2$; $\dim X \leq 3$	degree $2p$ with $2 \leq p \leq n - 2$
Hyperplane sections	reduction to middle degree, given the Lefschetz standard conjecture	the Lefschetz standard conjecture
Weil classes, deformation	the algebraic locus is dense and all or meagre; one semiregular cycle closes it	no semiregular cycle is known; Schoen's subvariety is undecided
Weil classes, degree bounds	equivalent to a uniform degree bound on a dense set	no such bound is known
Weil classes, Prym factors	for $\mathbb{Q}(\sqrt{-3})$, algebraic on A if and only if on a complement A'	no general complement is known; a count says none exists for $\dim A \geq 8$
Classes beyond Weil	algebraic at CM points of Mumford families	one Kuga–Satake class
Not of abelian type	every variety reached from abelian varieties; many hyper-Kähler varieties	the square of a K3 surface with real multiplication

TABLE 1. The routes and their stopping points.

Conventions. Varieties are smooth, complex, projective and connected unless said otherwise. For such an X we write $\mathrm{Hdg}^p(X) \subseteq H^{2p}(X, \mathbb{Q})$ for the Hodge classes and $\mathrm{Alg}^p(X) \subseteq \mathrm{Hdg}^p(X)$ for the algebraic ones, and $\mathrm{HC}(X, p)$ for the statement $\mathrm{Alg}^p(X) = \mathrm{Hdg}^p(X)$. We write $\mathrm{HC}(X)$ when this holds for every p .

2. REDUCTIONS THAT HOLD FOR EVERY VARIETY

2.1. Degree two.

Theorem 2.1 (Lefschetz [Lef24], Kodaira–Spencer [KS53]). *$\mathrm{HC}(X, 1)$ holds for every X .*

Proof. The exponential sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^\times \rightarrow 0$ gives an exact sequence

$$H^1(X, \mathcal{O}_X^\times) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \longrightarrow H^2(X, \mathcal{O}_X).$$

The second map is the composite of $H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathbb{C})$ with the projection onto $H^{0,2}(X)$ given by the Hodge decomposition. An integral class of type $(1, 1)$ has zero projection to $H^{0,2}$, so it is the first Chern class of a holomorphic line bundle. By the GAGA theorem of Serre that line bundle is algebraic, and its first Chern class is the class of a divisor. A rational Hodge class has an integral multiple, which is therefore the class of a divisor. $\square$

2.2. Hard Lefschetz. Let $h \in H^2(X, \mathbb{Q})$ be the class of a hyperplane section, and L the operator of cup product with h .

Proposition 2.2. *For $p < n/2$, $\mathrm{HC}(X, p)$ implies $\mathrm{HC}(X, n - p)$.*

Proof. By the hard Lefschetz theorem $L^{n-2p}: H^{2p}(X, \mathbb{Q}) \rightarrow H^{2n-2p}(X, \mathbb{Q})$ is an isomorphism. It maps $H^{p,p}$ to $H^{n-p, n-p}$, so it maps $\mathrm{Hdg}^p(X)$ onto $\mathrm{Hdg}^{n-p}(X)$: if β is a Hodge class of degree $2n-2p$ and $\beta = L^{n-2p}\alpha$, then L^{n-2p} maps the (r, s) components of α to the $(r+n-2p, s+n-2p)$ components of β , so α has type (p, p) . Cup product with h sends the class of a subvariety Z to the class of $Z \cap H$ for a general hyperplane H , so L^{n-2p} maps $\mathrm{Alg}^p(X)$ into $\mathrm{Alg}^{n-p}(X)$. Hence $\mathrm{Hdg}^{n-p}(X) = L^{n-2p} \mathrm{Hdg}^p(X) = L^{n-2p} \mathrm{Alg}^p(X) \subseteq \mathrm{Alg}^{n-p}(X)$. $\square$

Corollary 2.3. *$\mathrm{HC}(X, p)$ holds for $p \in \{0, 1, n-1, n\}$. In particular $\mathrm{HC}(X)$ holds when $\dim X \leq 3$.*

Proof. For $p = 0$ and $p = n$ the Hodge classes are multiples of the fundamental class and of the class of a point. Theorem 2.1 gives $p = 1$, and Proposition 2.2 with $p = 1$ gives $p = n - 1$. When $n \leq 3$ every p is one of these. $\square$

2.3. Descent.

Proposition 2.4. *Let $f: Y \rightarrow X$ be a surjective morphism of smooth projective varieties, and $r = \dim Y - \dim X$. Then $\mathrm{HC}(Y, p+r)$ implies $\mathrm{HC}(X, p)$. In particular $\mathrm{HC}(Y)$ implies $\mathrm{HC}(X)$.*

Proof. Let h_Y be the class of a hyperplane section of Y . The push-forward $f_*(h_Y^r) \in H^0(X, \mathbb{Q})$ is the degree d of a general fibre of f against h_Y^r , which is a positive integer. For $\alpha \in \mathrm{Hdg}^p(X)$ the class $f^*\alpha \cup h_Y^r$ is a Hodge class of degree $2(p+r)$ on Y , hence algebraic, and by the projection formula $f_*(f^*\alpha \cup h_Y^r) = \alpha \cup f_*(h_Y^r) = d\alpha$. Push-forward of cycles is compatible with the cycle class, so $d\alpha$, and hence α , is algebraic. $\square$

Taking $Y = X \times \mathbb{P}^1$ shows that the conjecture for $X \times \mathbb{P}^1$ gives it for X . This is the reason, explained in [BhH8], that the conjecture for the varieties that are not abelian already contains the conjecture for abelian varieties.

2.4. The reduction to middle degree. The remaining degrees are $2 \leq p \leq n - 2$, and by Proposition 2.2 it is enough to treat $2 \leq p < n/2$ and, when n is even, $p = n/2$. The natural way to move from degree $2p$ on X to the middle degree of a smaller variety goes through hyperplane sections, and it needs one more statement. Recall that the *Lefschetz standard conjecture* $B(X)$ asks that the inverse of L^{n-2p} , defined on $H^{2n-2p}(X, \mathbb{Q})$, be induced by an algebraic cycle on $X \times X$ for every p .

Proposition 2.5. *Let $2p < n$ and let $i: Y \hookrightarrow X$ be a smooth complete intersection of $n - 2p$ hyperplane sections. If $B(X)$ and $\mathrm{HC}(Y, p)$ hold, then $\mathrm{HC}(X, p)$ holds.*

Proof. Let $\alpha \in \mathrm{Hdg}^p(X)$. Then $i^*\alpha \in \mathrm{Hdg}^p(Y)$ is algebraic, say $i^*\alpha = \mathrm{cl}(Z)$, and $i_*i^*\alpha = \alpha \cup [Y] = L^{n-2p}\alpha$, so $L^{n-2p}\alpha = \mathrm{cl}(i_*Z)$ is algebraic. The inverse of L^{n-2p} is induced by an algebraic correspondence, which maps algebraic classes to algebraic classes, so α is algebraic. $\square$

The Lefschetz standard conjecture is itself open. It is known for curves and surfaces, for abelian varieties by Lieberman [Lie68, Kle68], and for hyper-Kähler varieties of $K3^{[n]}$ type by Charles and Markman [CM13]. It follows from the Hodge conjecture, so Proposition 2.5 is a reduction and not a proof. Without it we do not know how to get from degree $2p$ on X to any smaller variety, and this is the first place where the general argument stops.

2.5. What any proof must use. Two counterexamples constrain the shape of a proof. The integral version of the conjecture fails: Atiyah and Hirzebruch found torsion classes that are not algebraic [AH62], and Kollár found non-torsion integral Hodge classes with no algebraic representative although a multiple has one [Kol92]. So a proof has to use rational coefficients in an essential way. The version for compact Kähler manifolds fails too: Voisin found complex tori with Hodge classes that are not combinations of Chern classes of coherent sheaves [Voi02b]. So a proof has to use projectivity, and an argument that would work on every compact Kähler manifold cannot be right. Both facts are used below: Voisin's tori are what rule out one form of the deformation argument in Section 5.

3. THE SHAPE OF WHAT REMAINS

The following two statements were isolated in [BhH8, Section 14].

Definition 3.1. (i) (F2) For every abelian variety A , the Hodge classes of A that lie outside the subring generated by divisor classes and the Weil classes of quotients of A are algebraic.
(ii) (F3') For every X , every Hodge class on X is algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences.

Theorem 3.2 ([BhH8, Theorem 14.1(iv), Propositions on (F2) and (F3')]). (a) *The Hodge conjecture is equivalent to the conjunction of (F2) and (F3').*
(b) *(F2) is equivalent to the Hodge conjecture for abelian varieties.*
(c) *Over the implications proved in [BhH8] and in the literature, the minimal sets of statements that imply the conjecture are $\{F3\}$, $\{L, M\}$, $\{L, F3'\}$, $\{V, F3'\}$ and $\{F2, F3'\}$, where (F3) is the conjecture for varieties that are not abelian, L is the Lefschetz standard conjecture for every variety, M says that every Hodge class is motivated in the sense of André [And96], and V is the variational Hodge conjecture for algebraic classes.*

Part (a) has a simple meaning. Given (F2), every Hodge class on an abelian variety is algebraic, and an algebraic correspondence carries algebraic classes to algebraic classes; so (F3') then says that every Hodge class on every variety is algebraic. The attempt therefore splits into two halves: the conjecture for abelian varieties, which is (F2) and is the subject of Sections 4 to 6, and the conjecture modulo abelian varieties, which is (F3') and is the subject of Section 7. We note that (F3) alone is the whole conjecture, by Proposition 2.4 applied to $A \times \mathbb{P}^1 \rightarrow A$, so the split is a real division of the problem and not a reformulation of it.

4. ABELIAN VARIETIES

Let A be an abelian variety of dimension g and $V = H^1(A, \mathbb{Q})$. Then $H^*(A, \mathbb{Q}) = \bigwedge^* V^\vee$, and the Hodge structure on V is given by a homomorphism $h: \mathrm{U}(1) \rightarrow \mathrm{GL}(V_\mathbb{R})$. The *Hodge group* $\mathrm{Hg}(A)$ is the smallest algebraic subgroup of $\mathrm{GL}(V)$ defined over $\mathbb{Q}$ whose real points contain the image of h .

Proposition 4.1 ([BhH8, Section 4], [Del82]). *The Hodge classes of A in every degree are exactly the elements of $\bigwedge^* V^\vee$ fixed by $\mathrm{Hg}(A)$.*

The *Lefschetz group* of A is the centraliser of $\mathrm{End}^0(A)$ in the symplectic group of a polarization. Its invariants are generated by divisor classes, and the Hodge group is contained in it. When the two are equal, every Hodge class is a polynomial in divisor classes, and is algebraic. The Hodge classes that are not generated by divisors exist exactly when the Hodge group is strictly smaller than the Lefschetz group, and they come in two kinds.

The first kind are Weil's classes [Wei77]. Let K be an imaginary quadratic field acting on A of dimension $2n$ so that both embeddings of K occur with multiplicity n on $H^{1,0}(A)$. Then $W = \bigwedge_K^{2n} V^\vee$ is a $\mathbb{Q}$ -plane of Hodge classes in $H^{2n}(A, \mathbb{Q})$, the *Weil plane*, and for a general such A it is not generated by divisors. These are the subject of Section 5.

The second kind appear on abelian varieties whose Hodge group is smaller than the Lefschetz group for a reason other than a K -action. The simplest are Mumford's fourfolds [Mum69]: abelian fourfolds with $\text{End}(A) = \mathbb{Z}$ whose Hodge group is a form of SL_2^3 acting on V through the tensor product of the three standard representations. They have no exceptional Hodge classes on A itself, but $A \times A$ has two. These are the subject of Section 6.

The known cases of the conjecture for abelian varieties are those where the Hodge ring is generated by divisors (for instance when $\dim A \leq 3$, or when A is a power of an elliptic curve without complex multiplication), the fourfolds and some sixfolds of Weil type listed in Section 5, and the products and powers covered by [MZ99], [RM08] and [Fl25].

5. WEIL CLASSES

5.1. The reduction to one class. Fix an imaginary quadratic field K , an integer $n \geq 2$ and a nondegenerate K -hermitian form of signature (n, n) , with discriminant δ . The abelian varieties of Weil type with these data form a family Sh over a connected base whose universal cover is the bounded symmetric domain of $\text{SU}(n, n)$, of dimension n^2 . A very general member has Hodge group $\text{SU}(V, H)$, and for such a member every Hodge class is a polynomial in the polarization class η except in degree $2n$, where

$$\text{Hdg}^n(A) = \mathbb{Q}\eta^n \oplus W$$

[BhH8], [Wei77, Theorem 4]. Multiplication by a suitable element of K moves one nonzero Weil class onto a second one that spans the Weil plane with it, so on such a member the whole conjecture is the algebraicity of a single Weil class.

5.2. The algebraic locus. Let $\Sigma \subseteq \text{Sh}$ be the set of members on which the Weil class is algebraic. The following is proved in [BhH8] and in the companion note [BhW]; we list it to fix what the deformation route already has.

Theorem 5.1 ([BhH8, BhW]). *(a) Σ is a countable union of closed algebraic subsets of Sh , it is dense, and it is either all of Sh or meagre.*

- (b) $\Sigma = \text{Sh}$ if and only if the Weil class has algebraic representatives of bounded degree on a Zariski-dense set of members.*
- (c) If at one member there is a local complete intersection, or a perfect complex, whose class (respectively Chern character) in degree $2n$ is a nonzero Weil class plus a multiple of η^n , with higher terms in the span of the powers of η , and whose semiregularity map is injective, then $\Sigma = \text{Sh}$.*
- (d) No object whose Chern character in degree $2n$ is a pure Weil class is semiregular.*
- (e) At a member with Hodge group $\text{SU}(V, H)$, no cycle or complex built from divisors and homomorphisms by the usual operations represents a Weil class, and neither does any tautological cycle of a curve with an automorphism of order three.*

Part (d) is where the Kähler counterexample enters: a semiregular object would deform with the class, and a pure Weil class stays of Hodge type on a family of complex tori that contains Voisin's tori, which carry no subvarieties and no coherent sheaves with the required Chern character. Part (c) does not meet this obstruction, because η^n stops being of Hodge type as soon as the torus leaves the polarized family. So the deformation route asks for a semiregular object at some member whose class is a nonzero Weil class plus a multiple of η^n . Part (e)

says that such an object cannot be built from the obvious ingredients at a member with full Hodge group. At special members it can be, but transporting it to the general member by Hecke correspondences makes its degree grow without bound, and by part (b) bounded degree is exactly what is needed.

5.3. The known cases. The Weil class is known to be algebraic on every member in the following families: Weil fourfolds for $K = \mathbb{Q}(\sqrt{-3})$ of every discriminant [Sch88, Sch98]; a family of Weil sixfolds for $\mathbb{Q}(\sqrt{-3})$ [Sch98] and one for $\mathbb{Q}(\sqrt{-1})$ [Koi04], both obtained from Prym varieties of cyclic covers of curves of genus four; Weil fourfolds of discriminant one for every K , and their powers [Fl25]. Markman's preprints extend this to every Weil fourfold and every split Weil sixfold [Mar25a]. For $2n \geq 8$ no family is known in which the Weil class is algebraic on every member.

5.4. Schoen's cycles as one subvariety. The only cycles known to represent Weil classes at members with full Hodge group in dimension $2n \geq 8$ are Schoen's. We recall his construction and then extract from it a single subvariety of the shape that Theorem 5.1(c) asks for.

Let X be a curve of genus $g = n + 1 \geq 3$, let $L \in \text{Pic}^0(X)$ have exact order three, and let $\pi: C \rightarrow X$ be the étale cyclic triple cover defined by L , with covering automorphism σ . The curve C has genus $3g - 2$. In the Jacobian $J = J(C)$ let $N = 1 + \sigma + \sigma^2$ and $P = 3 - N \in \text{End}(J)$, and let $B = P(J)$, the *Prym variety* of the cover. Then $N^2 = 3N$, so P is multiplication by 3 on B , and σ acts on B with minimal polynomial $x^2 + x + 1$, which makes B an abelian variety of dimension $2n$ with an action of $K = \mathbb{Q}(\zeta)$, $\zeta = e^{2\pi i/3}$. Its hermitian form is split [Loo97], and both embeddings of K occur with multiplicity n on $H^{1,0}(B)$ [Sch88, Lemma 1.6a], so B is of Weil type. For very general (X, L) its Hodge group is $\text{SU}(V, H)$, by Looijenga's computation of the monodromy [Loo97] and André's theorem [And92].

Put $h = 2g - 2 = 2n$, fix $c_0 \in C$, and let

$$\Psi: C^h \rightarrow B, \quad (c_1, \dots, c_h) \mapsto P\left(\sum_i (c_i - c_0)\right).$$

Let $\Lambda \subseteq C^h$ be the set of h -tuples with $\pi(c_1) + \dots + \pi(c_h) \in |K_X|$. Since $|K_X| \cong \mathbb{P}^n$ and the map $C^h \rightarrow X^{(h)}$ is finite, Λ has pure dimension n .

Schoen's construction [Sch88, Section 2, case $r = 0$] runs as follows. Let $G' \subseteq \text{Aut}(C^h)$ be generated by the permutations of the factors and by the maps $\sigma^{v_1} \times \dots \times \sigma^{v_h}$ with $\sum v_i \equiv 0 \pmod{3}$, and let $W_0 = C^h/G'$ with quotient map ρ . The map $W_0 \rightarrow X^{(h)}$ is an étale cover of degree three, and over $|K_X|$, which is simply connected, it splits into three components Q_0, Q_1, Q_2 , permuted by $\delta = \sigma \times 1 \times \dots \times 1$. For each nontrivial character χ of $\mathbb{Z}/3$ the cycle $z_\chi = \sum_t \chi(-t) \delta_*^t Q_0$ has nonzero class [Sch88, proof of Theorem 2.0]. Patel and Zhang reach the same three components by base change of the Abel–Jacobi map along an isogeny onto $J(X)$ [PZ25, Theorem 1.1]. Let $\Lambda_0 = \rho^{-1}(Q_0) \subseteq \Lambda$.

Theorem 5.2. *In this situation:*

- (a) *the class $\Psi_*[\Lambda_0] \in H^{2n}(B, \mathbb{Q})$ pairs nontrivially with the Weil plane of B ;*
- (b) *there is an irreducible component Λ' of Λ_0 that maps generically finitely onto an n -dimensional subvariety $Y = \Psi(\Lambda')$ of B whose class pairs nontrivially with the Weil plane;*
- (c) *if $\text{Hg}(B) = \text{SU}(V, H)$, which holds for very general (X, L) , then*

$$\text{cl}(Y) = c\eta^n + w, \quad c \in \mathbb{Q}_{>0}, \quad 0 \neq w \in W.$$

Proof. We work with complex coefficients and write $\bar{\chi}$ for the conjugate character. Write $H^1(B, \mathbb{C}) = V_\chi \oplus V_{\bar{\chi}}$ for the eigenspaces of σ^* , each of dimension h . The Weil plane, tensored with $\mathbb{C}$, is $\bigwedge^h V_\chi \oplus \bigwedge^h V_{\bar{\chi}}$.

Step 1: the pull-back of the Weil plane. Write $\Psi = P' \circ \Xi$, where $\Xi: C^h \rightarrow J$ is $(c_i) \mapsto \sum (c_i - c_0)$ and $P': J \rightarrow B$ is P with its target restricted. If $\iota: B \rightarrow J$ is the inclusion, then $P'\iota$ is multiplication by 3 on B , so P'^* is injective on H^1 , and it carries V_χ into the χ -eigenspace of $H^1(C, \mathbb{C})$, which Ξ^* identifies with a summand of $H^1(C^h, \mathbb{C})$ on each factor. For a basis $y_1, \dots, y_h$ of V_χ let $\tilde{y}_j \in H^1(C, \mathbb{C})_\chi$ be the corresponding classes on C . Then

$$u_\chi := \Psi^*(y_1 \wedge \dots \wedge y_h) = \prod_{j=1}^h \left(\sum_{i=1}^h p_i^* \tilde{y}_j \right),$$

where p_i is the i -th projection. A product $\tilde{y}_j \cup \tilde{y}_k$ lies in the χ^2 -eigenspace of $H^2(C, \mathbb{C})$. Since σ acts trivially on $H^2(C, \mathbb{C})$ and $\chi^2 \neq 1$, that eigenspace is zero, so the product vanishes. Expanding the product, only the terms with one factor $p_i^* \tilde{y}_j$ for each i survive, and $u_\chi = \sum_s \pm \prod_i p_i^* \tilde{y}_{s(i)}$, the sum over permutations s . By the Künneth formula $u_\chi \neq 0$.

From here on automorphisms act on cohomology by pull-back. The class u_χ is invariant under G' : the permutations fix it because Ψ is symmetric, and each $\sigma^{v_1} \times \dots \times \sigma^{v_h}$ multiplies it by $\chi(\sum v_i) = 1$. The map δ normalizes G' , and δ^* multiplies u_χ by $\chi(1)$. Let U_χ be the part of $H^h(C^h, \mathbb{C})^{G'}$ on which δ^* acts by $\chi(1)$. It is one-dimensional [Sch88, Lemma 1.2]. (Directly: a Künneth product of eigenclasses is fixed by every $\sigma^{v_1} \times \dots \times \sigma^{v_h}$ with $\sum v_i \equiv 0$ only if all its factors lie in one eigenspace of σ^* , and δ^* then acts by $\chi(1)$ only if that eigenspace is $H^1(C, \mathbb{C})_\chi$, which has dimension $2g - 2 = h$; the symmetric part is its top exterior power.) So u_χ spans U_χ . In the same way $u_{\bar{\chi}} = \Psi^*(\bar{y}_1 \wedge \dots \wedge \bar{y}_h)$ spans $U_{\bar{\chi}}$.

Step 2: Ψ is generically finite. At a point $(c_1, \dots, c_h)$ the differential of Ψ is dual to the map that sends a holomorphic 1-form ω on B , viewed as a form on C in the nontrivial eigenspaces, to $(\omega(c_1), \dots, \omega(c_h))$. These forms make up a space of dimension $2n = h$, and h general points impose independent conditions on it, so the differential is an isomorphism at a general point. Hence Ψ is surjective and generically finite, of some degree $e > 0$, and

$$\int_{C^h} u_\chi \cup u_{\bar{\chi}} = e \int_B (y_1 \wedge \dots \wedge y_h) \wedge (\bar{y}_1 \wedge \dots \wedge \bar{y}_h) \neq 0.$$

Step 3: the pairing. The class $\lambda_0 = [\Lambda_0] = \rho^*[Q_0]$ is G' -invariant; ρ is unramified at the general point of Λ_0 , since G' acts freely on tuples whose images in X are distinct. Since $\delta^* = (\delta^{-1})_*$, the component of λ_0 on which δ^* acts by $\chi(1)$ is

$$\frac{1}{3} \sum_t \chi(t) \delta_*^t \lambda_0 = \frac{1}{3} \rho^*[z_{\bar{\chi}}],$$

which is nonzero by Schoen's theorem applied to $\bar{\chi}$, because ρ^* is injective in rational cohomology. It lies in $U_{\bar{\chi}}$, so it equals $a u_{\bar{\chi}}$ with $a \neq 0$. Because δ preserves the intersection pairing, a class on which δ^* acts by $\psi(1)$ pairs to zero with $u_{\bar{\chi}}$ unless $\psi = \chi$. Hence

$$\int_{C^h} \lambda_0 \cup u_{\bar{\chi}} = a \int_{C^h} u_\chi \cup u_{\bar{\chi}} \neq 0,$$

and by the projection formula $\int_B \Psi_* \lambda_0 \cup (\bar{y}_1 \wedge \dots \wedge \bar{y}_h) \neq 0$. This is (a).

Step 4: one component. The class $\Psi_* \lambda_0$ is $\sum_k e_k \text{cl}(\Psi(\Lambda_k))$, the sum over the irreducible components Λ_k of Λ_0 on which Ψ is generically finite, with e_k the degree of $\Lambda_k \rightarrow \Psi(\Lambda_k)$; the other components contribute nothing. By (a) one term pairs nontrivially with the Weil plane. This is (b).

Step 5: the shape of the class. If $\mathrm{Hg}(B) = \mathrm{SU}(V, H)$ then $\mathrm{cl}(Y) \in \mathrm{Hdg}^n(B) = \mathbb{Q}\eta^n \oplus W$, so $\mathrm{cl}(Y) = c\eta^n + w$. The centre $\mathrm{U}(1) \subseteq \mathrm{U}(V, H)$, acting by scalars of $K \otimes \mathbb{R} = \mathbb{C}$ of absolute value one, fixes η and the orientation class, and acts on the two lines of $W \otimes \mathbb{C}$ by $\lambda^{\pm 2n}$. So $\int \eta^n \cup w' = 0$ for every $w' \in W$. By (b), $w \neq 0$. Finally $\int \mathrm{cl}(Y) \cup \eta^n$ is the degree of Y with respect to the ample class η , which is positive, and it equals $c \int \eta^{2n}$, so $c > 0$. $\square$

Corollary 5.3. *Let $n \geq 4$. Suppose that for one very general (X, L) of genus $n+1$ the subvariety Y of Theorem 5.2, or its structure sheaf as a perfect complex on B , is semiregular. Then the Weil classes are algebraic on every member of the split Weil family of $\mathbb{Q}(\sqrt{-3})$ in dimension $2n$.*

Proof. If Y is a local complete intersection, its class is $c\eta^n + w$ with $w \neq 0$, which is the shape required by Theorem 5.1(c). For the structure sheaf, $\mathrm{ch}(\mathcal{O}_Y)$ has $\mathrm{cl}(Y)$ in degree $2n$ and Hodge classes in higher degrees, which at a member with Hodge group $\mathrm{SU}(V, H)$ are multiples of powers of η by the description of the Hodge ring. In either case Theorem 5.1(c) gives $\Sigma = \mathrm{Sh}$. $\square$

5.5. Where the Weil route stops. Everything now rests on one question.

Question 5.4. For $n \geq 4$ and very general (X, L) , is the subvariety $Y \subseteq B$ of Theorem 5.2 semiregular? Equivalently, at first order: does Y deform along every one of the n^2 directions of the Weil family at B , and not only along the $3n$ directions that come from deforming (X, L) ?

We have not been able to decide it, and the closest analogue points the wrong way. The Abel–Jacobi curve $C \subseteq J(C)$ is built from the same map as Y . Its class $\theta^{g-1}/(g-1)!$ is algebraic on every principally polarized abelian variety, so it stays of Hodge type everywhere. Yet by the criterion of Matsusaka and Ran [Mat59, Ran81] a curve with this class exists only on Jacobians and on products of Jacobians, which form a proper subvariety of the moduli space once $g \geq 4$. By Bloch’s theorem [Blo72] the Abel–Jacobi curve is therefore not semiregular. Schoen’s subvariety is the analogue one step up, built from the fibre of the Abel–Jacobi map over the canonical class, and nothing in its construction moves it off the locus of Prym varieties. This is an analogy and not a proof. If the answer to Question 5.4 is no, the deformation route needs a different object, and Theorem 5.1(e) says it cannot come from divisors, homomorphisms or the tautological cycles of curves.

The theorem also sits consistently against the barrier of [BhW]. Since $3 - (1 + \sigma + \sigma^2) = (1 - \sigma)(2 + \sigma)$, the map Ψ is $(c_i) \mapsto (1 - \sigma) \sum_i (2 + \sigma)(c_i - c_0)$, which has the form of the maps in [BhW, Proposition 11]; and $(1 - \sigma)(J) = B$, because $(2 + \sigma)(2 + \sigma^2) = 3$ on $(1 - \sigma)(J)$, so that $2 + \sigma$ is an isogeny of it. That proposition says that Ψ_* carries every class in the subring of $H^*(C^h, \mathbb{Q})$ generated by the diagonals, the graphs of σ and σ^2 and the point classes to a class invariant under the unitary group, and such a class pairs to zero with the Weil plane. By Theorem 5.2(a), then, $[\Lambda_0]$ does not lie in that subring. This is what the barrier predicts: a Weil class at a Prym variety with full Hodge group can come only from a cycle outside the tautological ring, and Schoen’s splitting of the canonical system into three components produces one.

For fields other than $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-1})$, and for nonsplit forms when $2n \geq 6$, no cycle is known at any member with full Hodge group, so the route has nothing to deform.

6. CLASSES BEYOND THE WEIL CLASSES

By Theorem 3.2(b), the conjecture for abelian varieties also needs the classes that divisors and Weil classes do not generate. Their first instance is Mumford’s.

Let A be a Mumford fourfold, so that $\mathrm{End}(A) = \mathbb{Z}$ and $\mathrm{Hg}(A)$ is a $\mathbb{Q}$ -form of SL_2^3 acting on $V_{\mathbb{C}} = S_1 \otimes S_2 \otimes S_3$. These fourfolds form families over Shimura curves [Mum69]. The space $\bigwedge^2 V_{\mathbb{C}}$

contains $\mathrm{Sym}^2 S_1 \otimes \mathrm{Sym}^2 S_2 \otimes \mathbf{1}$ and its two companions, and on $A \times A$ the invariants of $\mathrm{Hg}(A)$ include, besides the classes generated by divisors, the two classes that come from the identity of each of these three pieces modulo the polarization. They are Hodge classes that are neither Lefschetz nor Weil classes. For them [BhH8] proves: they are algebraic at every CM point of the family; they are algebraic on every member once one Kuga–Satake class, on a K3 surface of Picard number thirteen attached to the fourfold, is algebraic; and they are algebraic on every member exactly when the Lefschetz standard conjecture holds for one ninefold. It also shows that no construction from line bundles by cones, tensor products, pull-backs, push-forwards and Fourier–Mukai functors reaches them, because all of these preserve invariance under the Lefschetz group.

Here the attempt stops at one class. The Kuga–Satake correspondence is known to be algebraic for a few families of K3 surfaces (see Section 7), and the K3 surfaces attached to Mumford’s fourfolds are not among them.

7. VARIETIES THAT ARE NOT OF ABELIAN TYPE

The statement (F3′) holds trivially on every variety whose cohomology is reached from abelian varieties by algebraic correspondences, and Proposition 2.4 spreads it along surjective morphisms. The literature adds a good deal.

- *Dimension at most three, and codegrees 0 and 2* hold by Corollary 2.3.
- *K3 surfaces of Picard number 19 or 20* carry a Shioda–Inose structure, which relates their transcendental lattice to that of an abelian surface [Mor84].
- *Products of surfaces.* The conjecture holds for every product of surfaces with $q = 2$ and $p_g = 1$, and for every self-product of a K3 surface whose field of Hodge endomorphisms of the transcendental lattice is a CM field [RM08]. It holds for every power of a K3 surface associated with a variety of generalized Kummer type [Fl25], and for the squares of the K3 surfaces in an explicit family covered by squares of curves of genus seven [ILP21].
- *Hyper-Kähler varieties.* Rational Hodge isometries between varieties of $K3^{[n]}$ type are algebraic [Mar22], and the standard conjectures hold for them [CM13]. The Chow motive of a generalized Kummer variety lies in the tensor category generated by the abelian surface, so the conjecture holds for all products of generalized Kummer varieties [Xu15, FTV19]. The same holds for O’Grady’s six-dimensional varieties [Fl23], and O’Grady’s ten-dimensional varieties have motives built from those of K3 surfaces [FFZ21]. The motive of a smooth moduli space of stable sheaves on a K3 or abelian surface S is a summand of a sum of twisted motives of powers of S [Bül18].
- *Fourfolds.* The conjecture holds for uniruled fourfolds [CM78] and for cubic fourfolds [Zuc77], and cubic fourfolds whose lattice of algebraic 2-cycles has rank greater than 19 have abelian motive [ABLP20]. Every smooth Delsarte hypersurface, and every cyclic cover branched along one, satisfies (F3′) [BhH8].

The first variety on which (F3′) is open is the square $S \times S$ of a K3 surface S whose transcendental lattice has real multiplication by a totally real field $E \neq \mathbb{Q}$ [BhH8]. The endomorphisms in E give Hodge classes on $S \times S$ in $H^2(S) \otimes H^2(S)$ that are not generated by divisors. If the Kuga–Satake correspondence between S and its Kuga–Satake abelian variety were algebraic, these classes would move to Hodge classes on the square of an abelian variety, and (F3′) for $S \times S$ would become a case of (F2). It is not known to be algebraic for such S . The analysis of [BhH8] gives the same structure here as for Weil classes: the algebraic locus of the real multiplication in a family of such K3 surfaces is dense, contains every CM point, and is either the whole family or meagre. Deciding between the two is open.

Beyond the K3 surfaces lies the general case of (F3'): a variety of dimension at least four whose motive is not known to come from abelian varieties, with Hodge classes in a degree that Corollary 2.3 does not reach. The methods that have produced cycles in the known cases are few: divisors, through Theorem 2.1; rational curves, as in [CM78]; normal functions over Lefschetz pencils, as for cubic fourfolds [Zuc77]; the inductive structure of Fermat and Delsarte varieties; and correspondences with curves, abelian varieties and moduli of sheaves, as in the results just listed. We do not see how to apply any of them to a variety with none of this structure, and that is where the attempt at (F3') stops.

8. SPREADING SCHOEN'S CONSTRUCTION

Schoen's cycles reach a family of dimension $3n$ inside one of dimension n^2 , and Question 5.4 asks whether they can be deformed out of it. This section tries the two other ways of spreading them: carrying them to other abelian varieties by correspondences, and running the construction on other varieties. The first gives a theorem. It makes the Weil class of any abelian variety of Weil type for $\mathbb{Q}(\sqrt{-3})$ algebraic exactly when the Weil class of a second, complementary variety is, but a count of dimensions shows that it cannot by itself reach the general member. The second reaches neither (F2) nor (F3'), for reasons we make precise.

Throughout the section $K = \mathbb{Q}(\sqrt{-3})$ and ζ is a primitive cube root of unity. Varieties of Weil type are for K , with any discriminant, and we assume that $\mathcal{O}_K$ acts, which can always be arranged by passing to a K -isogenous variety. For such a variety A of dimension $2m$ write $W(A) \subseteq H^{2m}(A, \mathbb{Q})$ for its Weil plane. Over $\mathbb{C}$ the space $H^1(A, \mathbb{C})$ is the sum of the eigenspaces $V_\chi(A)$ and $V_{\bar{\chi}}(A)$ of ζ^* , each of dimension $2m$, and $W(A) \otimes \mathbb{C} = \bigwedge^{2m} V_\chi(A) \oplus \bigwedge^{2m} V_{\bar{\chi}}(A)$. We call A *Weil-algebraic* if every class in $W(A)$ is algebraic. One nonzero algebraic class in $W(A)$ is enough, because a suitable element of $\mathcal{O}_K$ acting as an endomorphism moves it to a second class that spans $W(A)$ with it [BhW, proof of Theorem 6].

8.1. Products and complements.

Lemma 8.1. *Let B , A_1 and A_2 be of Weil type, of dimensions $2(m_1 + m_2)$, $2m_1$ and $2m_2$, and let $f: B \rightarrow A_1 \times A_2$ be an isogeny that commutes with $\mathcal{O}_K$.*

- (a) *The isomorphism f^* carries a plane in $W(A_1) \otimes W(A_2) \subseteq H^{2m_1}(A_1) \otimes H^{2m_2}(A_2)$ onto $W(B)$. Over $\mathbb{C}$ this plane is spanned by $\omega_1 \otimes \omega_2$ and $\bar{\omega}_1 \otimes \bar{\omega}_2$, where ω_i spans $\bigwedge^{2m_i} V_\chi(A_i)$ and $\bar{\omega}_i$ is its complex conjugate.*
- (b) *If two of B , A_1 and A_2 are Weil-algebraic, so is the third.*

Proof. The map f^* is an isomorphism on rational cohomology that commutes with $\mathcal{O}_K$ and carries algebraic classes to algebraic classes, and so does its inverse $(\deg f)^{-1}f_*$. So we may take $B = A_1 \times A_2$ with the diagonal action. Then $V_\chi(B) = V_\chi(A_1) \oplus V_\chi(A_2)$, and the top exterior power of a direct sum is the tensor product of the top exterior powers. This is (a).

If A_1 and A_2 are Weil-algebraic, every class in $W(A_1) \otimes W(A_2)$ is a sum of products $\text{pr}_1^* \alpha_1 \cup \text{pr}_2^* \alpha_2$ of algebraic classes, so B is Weil-algebraic by (a). Suppose now that B and A_2 are Weil-algebraic. On $W(A_2)$ we have $\int \omega_2 \cup \omega_2 = 0$, because $\bigwedge^{4m_2} V_\chi(A_2) = 0$, while $p := \int \omega_2 \cup \bar{\omega}_2 \neq 0$, because $\bigwedge^{2m_2} V_\chi(A_2) \otimes \bigwedge^{2m_2} V_{\bar{\chi}}(A_2)$ is the top exterior power of $H^1(A_2, \mathbb{C})$. The number p is real, since ω_2 and $\bar{\omega}_2$ have even degree and so commute. Take nonzero rational classes $x \in W(B)$ and $\gamma \in W(A_2)$. Being real, they have the form $x = s\omega_1 \otimes \omega_2 + \bar{s}\bar{\omega}_1 \otimes \bar{\omega}_2$ and $\gamma = t\omega_2 + \bar{t}\bar{\omega}_2$ with $s, t \neq 0$. The class

$$\text{pr}_{1*}(x \cup \text{pr}_2^* \gamma) = p(s\bar{t}\omega_1 + \bar{s}t\bar{\omega}_1)$$

is rational, algebraic and nonzero, and it lies in $W(A_1)$. So A_1 is Weil-algebraic. $\square$

Theorem 8.2. *Let A be an abelian variety of Weil type for K , of dimension $2m \geq 2$, on which $\mathcal{O}_K$ acts. Then there are a smooth curve $C \subseteq A$ stable under ζ , on which ζ acts freely, so that $\pi: C \rightarrow X = C/\zeta$ is an étale cyclic triple cover of a curve of some genus $g \geq m+1$; a variety A' of Weil type of dimension $2(g-1-m)$; and an isogeny commuting with $\mathcal{O}_K$ between the Prym variety B of π and $A \times A'$. For every such curve, A is Weil-algebraic if and only if A' is.*

Proof. The curve. The automorphism ζ of A has finitely many fixed points, the kernel of the isogeny $1 - \zeta$. Let L be ample on A . The bundle $N = L \otimes \zeta^* L \otimes \zeta^{2*} L$ satisfies $\zeta^* N \cong N$, so, the group being cyclic, it carries a ζ -linearization, and the linearized bundle N^3 descends to an ample line bundle $\bar{M}$ on the normal projective quotient $q: A \rightarrow A/\zeta$, whose singular points are among the images of the fixed points. For k large, $2m-1$ general members of $|\bar{M}^k|$ meet in a smooth irreducible curve X that avoids those finitely many points, by Bertini's theorem. Then $C = q^{-1}(X)$ is étale over X , hence smooth, and it is the complete intersection of the $2m-1$ ample divisors $q^* D_i$, hence connected. So $\pi: C \rightarrow X$ is an étale cyclic triple cover, and ζ acts on C as its covering automorphism σ . Schoen observes the existence of such curves in [Sch88, Section 3].

The surjection. By the Lefschetz hyperplane theorem, restriction $H^1(A, \mathbb{Q}) \rightarrow H^1(C, \mathbb{Q})$ is injective. So the homomorphism $\alpha: J(C) \rightarrow A$ that adds up the points of a divisor of degree zero is surjective, and it commutes with ζ . In $\mathcal{O}_K$ we have $1 + \zeta + \zeta^2 = 0$, so $\alpha \circ (1 + \sigma + \sigma^2) = 0$ and $\alpha \circ P = 3\alpha$ for $P = 3 - (1 + \sigma + \sigma^2)$. Hence α maps the Prym variety $B = P(J(C))$ onto A , and in particular $g-1 \geq m$.

The complement. Let A' be the connected component of the kernel of $\alpha|_B$, and A'' its complement in B for the polarization induced from $J(C)$. Since σ preserves that polarization, both are stable under σ , so $\mathcal{O}_K$ acts on them. The maps $A'' \rightarrow A$ and $A' \times A'' \rightarrow B$ are isogenies commuting with $\mathcal{O}_K$, and composing one with a quasi-inverse of the other relates B and $A \times A'$.

Weil type. By [Sch88, Lemma 1.6a] the two eigenspaces of σ on $H^{1,0}(B)$ both have dimension $g-1$, and on $H^{1,0}(A)$ both have dimension m . So on $H^{1,0}(A')$ both have dimension $g-1-m$, and A' is of Weil type.

The equivalence. The cover is étale, so Schoen's Theorem 2.0 applies to it with $r=0$, and by [Sch88, Corollary 3.1] the variety B is Weil-algebraic. Lemma 8.1(b) gives the equivalence. $\square$

When $g-1=m$ the complement is zero and A is itself K -isogenous to a Prym variety; that happens on the Prym locus, of dimension $3m$, and nowhere else. For larger g the theorem trades the Weil class of A for that of A' , and since curves of higher degree give complements of any size, one can hope to choose the curve so that A' is a variety on which the Weil class is already known.

Corollary 8.3. *A is Weil-algebraic as soon as, for one curve as in Theorem 8.2, the complement A' is K -isogenous to a product of Weil-algebraic varieties. Among these are Prym varieties of étale cyclic triple covers; abelian surfaces of Weil type; Weil fourfolds for K [Sch88, Sch98]; and the varieties $E^k \times \bar{E}^k$, where E is an elliptic curve with complex multiplication by $\mathcal{O}_K$ and $\bar{E}$ is E with $\mathcal{O}_K$ acting through complex conjugation.*

Proof. Products of Weil-algebraic varieties are Weil-algebraic by Lemma 8.1(b), and Theorem 8.2 finishes the argument. On an abelian surface the Weil plane lies in H^2 and consists of Hodge classes, which are algebraic by Theorem 2.1. On $E^k \times \bar{E}^k$, with coordinates $z_1, \dots, z_{2k}$ on the factors, V_χ is spanned by $dz_1, \dots, dz_k$ and $d\bar{z}_{k+1}, \dots, d\bar{z}_{2k}$, so up to sign

$$\omega_\chi = \bigwedge_{i=1}^k (dz_i \wedge d\bar{z}_{k+i}).$$

Each factor is a class of type $(1, 1)$ on $E \times \bar{E}$, whose Néron–Severi group has rank $4 = h^{1,1}$ because E has complex multiplication, so it is a complex combination of divisor classes. Hence ω_χ and its conjugate are polynomials in divisor classes. $\square$

Remark 8.4. For a general A with $m \geq 4$ one cannot expect a complement of these kinds. Pryms of étale triple covers of curves of genus $m + k + 1$ form a family of dimension $3(m + k)$ inside the Weil family of dimension $(m + k)^2$. The varieties K -isogenous to $A \times A'$, with A in the family of dimension m^2 and A' in a family of dimension f , form a subvariety of dimension $m^2 + f$. For A to be general, the two must meet in dimension at least m^2 , which needs

$$3(m + k) + f \geq (m + k)^2$$

unless they meet in more than the expected dimension. Products of the varieties in Corollary 8.3 of total dimension $2k$ have at most $3k$ moduli, and $3m + 6k < (m + k)^2$ for every $k \geq 0$ once $m \geq 4$. For $m = 3$ only $k = 0$ passes, which is the Prym locus itself, and for $m = 2$ the values $k \leq 2$ pass. So, unless the Prym locus meets these product loci in excess dimension, the corollary reaches only a meagre subset of the family once $m \geq 4$, as Schoen anticipated [Sch88, Section 3]. Such excess intersections are what the Zilber–Pink philosophy says should not happen. This is a count and not a proof.

8.2. Other varieties, and (F2) and (F3'). *(F2).* Schoen’s classes are Weil classes. By [Sch88, Lemma 1.2 and Remark 1.4], for every curve with a cyclic group of automorphisms the space of classes his Theorem 2.0 makes algebraic is the pull-back of the Weil plane of a generalized Prym variety with cyclotomic multiplication. For cyclotomic fields of degree at least four this bears on the third missing input of [BhH8], the propagation for CM fields of higher degree. It does not bear on (F2). Schoen’s own count in [Sch88, Section 3] shows that these generalized Pryms can fill a whole Weil family in at most four small cases. To reach a class of (F2), such as the exceptional classes on $A \times A$ for a Mumford fourfold A , the construction would need an algebraic correspondence from a Prym-type variety B to $A \times A$ that carries a Weil class of B to the exceptional class. That correspondence is a Hodge class on $B \times A \times A$, so asking for it is again an instance of the conjecture for abelian varieties. Nor can the construction be run on A itself: a Mumford fourfold has $\text{End}(A) = \mathbb{Z}$, so it has no automorphism of order three.

(F3'). Schoen defines the same space of classes for any smooth projective C of odd dimension k with a cyclic group of automorphisms, under a few hypotheses, the main one being that the part V of $H^k(C, \mathbb{Q})$ on which the group acts by primitive characters has only two Hodge types. His example is a quintic threefold invariant under $\mathbb{Z}/5$ and missing the fixed points, where the space lies in $H^{120}(C^{40}, \mathbb{Q})$, has type $(60, 60)$, and was not known to be algebraic [Sch88, Section 1]. In that example $H^{3,0}$ is invariant, so V has types $(2, 1)$ and $(1, 2)$. It is then of level one, and so it is a Tate twist of H^1 of an abelian variety J_V , and the classes are images of the Weil plane of J_V under a Hodge isomorphism. They would be algebraic if J_V were Weil-algebraic and the isomorphism were induced by an algebraic correspondence. The second requirement is the generalized Hodge conjecture for the level-one part of $H^3(C)$, an instance of (F3') that we do not know to be proved for these quintics. The cycles of [Sch88, Section 2] come from the Abel–Jacobi map of a curve, which supplies that correspondence for free. For a variety of higher dimension nothing supplies it, so the construction stops at curves. When the two Hodge types are not adjacent, V is not of level one, and the construction has no abelian variety to work on.

So spreading Schoen’s construction by correspondences reduces the Weil class of A to that of a complement without making it easier, and spreading it to other varieties leads back to (F2) and (F3') themselves. Neither closes anything that Question 5.4 leaves open.

9. WHAT IS PROVED AND WHAT IS NOT

Proved here, with complete proofs or by citation of refereed work:

- (1) every Hodge class of degree 2 or $2n - 2$, and every Hodge class on a variety of dimension at most three, is algebraic;
- (2) the conjecture descends along surjective morphisms, and reduces to middle degree on complete intersections given the Lefschetz standard conjecture;
- (3) on every Prym variety B of an étale cyclic triple cover of a curve of genus $n + 1$, Schoen's construction gives an irreducible n -dimensional subvariety Y whose class pairs nontrivially with the Weil plane, and at a very general such B its class is $c\eta^n + w$ with $c > 0$ and w a nonzero Weil class (Theorem 5.2);
- (4) if that subvariety is semiregular for one very general cover of genus $n + 1$, the Weil classes are algebraic on every member of the split Weil family of $\mathbb{Q}(\sqrt{-3})$ in dimension $2n$ (Corollary 5.3);
- (5) every abelian variety A of Weil type for $\mathbb{Q}(\sqrt{-3})$ is, up to isogeny, a factor $A \times A'$ of the Prym variety of an étale cyclic triple cover, with A' of Weil type, and A is Weil-algebraic if and only if A' is (Theorem 8.2); in particular A is Weil-algebraic whenever A' is a product of the varieties listed in Corollary 8.3.

Recalled from [BhH8] and used as the frame: the conjecture is equivalent to (F2) and (F3') together; (F2) is the conjecture for abelian varieties; and the minimal sets of hypotheses are those of Theorem 3.2(c).

Not proved here, and open:

- (i) the Lefschetz standard conjecture, in general;
- (ii) the Weil classes on a general abelian variety of Weil type of dimension at least eight, and of dimension six outside the known families; in particular Question 5.4, and the existence, for a general A of dimension at least eight, of a complement as in Theorem 8.2 on which the Weil classes are known, which the count of Remark 8.4 makes unlikely;
- (iii) the exceptional classes on the square of a Mumford fourfold, which follow from one Kuga-Satake class;
- (iv) the conjecture for the square of a K3 surface with real multiplication, the first open case of (F3'), and (F3') in general.

So the attempt in full ends where it was always likely to end. The conjecture is equivalent to two statements, and each of them has an explicit first open case. The most concrete thing this paper adds is that the Weil case for $\mathbb{Q}(\sqrt{-3})$ in every dimension now hangs on a single, explicitly described subvariety of a Prym variety and the question whether it deforms.

Machine checks. No proof in this paper depends on a computer. The finite parts of the arguments have nevertheless been repeated by machine, and the files are in the author's repository [BhHC].

In Lean 4, checked by its kernel with nothing beyond the core library, we verified five things. The first is the deductions of Propositions 2.2, 2.4 and 2.5 and of Corollary 2.3, with every geometric input stated as a hypothesis, so that the dependence of Proposition 2.5 on the Lefschetz standard conjecture is visible in the statement. The second is the counts of genera and dimensions in Section 5, for every genus at once. The third is the arithmetic of the characters of $\mathbb{Z}/3$ in $\mathbb{Z}[\zeta]$ used in Step 3 of Theorem 5.2. The fourth is the weight count of Step 5 and the deduction of $c > 0$ and $w \neq 0$ from it. The identity $3 - (1 + \sigma + \sigma^2) = (1 - \sigma)(2 + \sigma)$ used after Question 5.4 is checked there too. The fifth, for Section 8, is the dimensions in Theorem 8.2, including the equality of the two eigenspaces on A' , the relation $\alpha \circ P = 3\alpha$, and the arithmetic

of Remark 8.4 for every m and k at once. It checks only that arithmetic; the remark itself remains a count and not a proof.

In Julia, and in a Python version of the same checks, all in exact arithmetic, we computed the following. On a general abelian variety of Weil type of dimension $2n \leq 8$, the Hodge classes span one line in each even degree except the middle one, where they span three. For a Weil class w the products $\eta \cup w$ and $\eta^n \cup w$ vanish. The space U_χ of Step 1 is one-dimensional for $g = 3$ and $g = 4$; for $g = 3$ this was checked by summing over the whole group. For $h \leq 6$ the class u_χ has $h!$ terms, is symmetric, and pairs nontrivially with $u_{\bar{\chi}}$. For Lemma 8.1 with $2m_1, 2m_2 \in \{2, 4\}$ we computed the push-forward of $x \cup \text{pr}_2^* \gamma$ in the exterior algebra and found the formula of the proof, and for Corollary 8.3 with $k \leq 4$ we checked that ω_χ is, up to sign, the product of the classes $dz_i \wedge d\bar{z}_{k+i}$.

None of this touches the geometry the paper relies on: the theorems of Schoen, Bloch, Looijenga and André, and those of [BhH8]. None of it bears on Question 5.4.

A note on preparation. Drafts of parts of this paper were prepared with the help of an AI language model, used under the author's direction. The author has checked every statement and proof and is responsible for the content.

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